# Contents

# Statistical analysis 1

## Introduction

Scientists use statistics to help them analyse and understand the evidence they collect during experiments. Statistics is an area of mathematics that measures variation in data, and the differences and relationships between sets of data. By using statistics, we can examine samples of populations or experimental results and decide how certain we can be about the conclusions we draw from them.

## 1.1 Mean and distribution

### Assessment statements

- State that error bars are a graphical representation of the variability of data.
- Calculate the mean and standard deviation of a set of values.
- State that the term 'standard deviation' is used to summarise the spread of values around the mean, and that 68% of all values fall within plus or minus one standard deviation of the mean.
- Explain how the standard deviation is useful for comparing the means and the spread of data between two or more samples.
- Deduce the significance of the difference between two sets of data using calculated values for *t* and the appropriate tables.
- Explain that the existence of a correlation does not establish that there is a causal relationship between two variables.

### Calculating the mean

The **mean** is an average of all the data that has been collected. For example, if you measure the height of all the students in your class, you could find the mean value by first calculating the sum of all the values and then dividing by the number of values.

$$\bar{x} = \frac{\Sigma x}{n}$$

### A normal distribution

If you measured the height of ten students and plotted the values on a graph, with the height on the *x*-axis and the number in each height group on the *y*-axis, you would get a result that did not show an obvious trend. If you measured the height of 100 students, your graph would begin to look bell shaped, with most values in the middle and fewer on either side. As you measured more and more students' heights, your graph would eventually become a smooth curve as in Figure **1.1** (overleaf). This bell-shaped graph of values is called a **normal distribution**.

Normal distribution curves may be tall and narrow, if all the values are close together, or flatter and wider when the data are more spread out. The mean value is at the peak of the curve.

**Mathematical vocabulary**

**You will find it helpful to understand some mathematical vocabulary first.**

- $x$ represents a single value – for example, a person's height $x = 1.7\,\text{m}$
- $n$ represents the total number of values in a set
- $\bar{x}$, called 'x bar', represents the mean of a set of values
- $\Sigma$ represents the sum of the values
- $s$ represents the standard deviation of the sample
- $\pm$ represents 'plus-or-minus'
- $t$ is a value calculated using a formula called the '*t*-test'

For example, Table 1.2 shows data collected on heart rate during exercise. A different value is recorded in each trial. For a small number of values (three or four), the mean is plotted and the error bar added to show the highest and lowest values, as in Figure 1.3. This shows the **range** of the values. For a larger number of values (five or more), the standard deviation is calculated and this is shown in the same way (Figure 1.4).

| Trial number | Heart rate/ beats $min^{-1}$ |
|---|---|
| 1 | 135 |
| 2 | 142 |
| 3 | 139 |
| mean | 139 |

**Table 1.2** For a small number of values, only the mean is calculated.

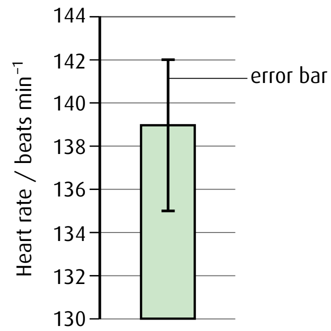



**Figure 1.3** Here, the error bar shows the range of the data. The mean value of 139 is plotted with the error bar ranging from the highest value, 142, to the lowest value, 135.

| Trial number | Heart rate/ beats $min^{-1}$ |
|---|---|
| 1 | 137 |
| 2 | 141 |
| 3 | 134 |
| 4 | 136 |
| 5 | 140 |
| 6 | 139 |
| mean | 138 |
| standard deviation | 2.6 |

**Table 1.3** For five or more values, the standard deviation is also calculated.

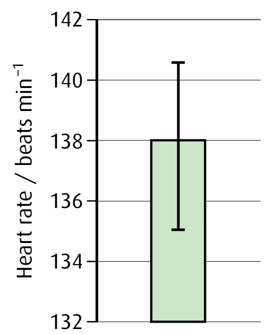



**Figure 1.4** This time, the error bar shows the standard deviation from the mean. The mean value of 138 is plotted with the error bar showing $\pm 1s$.

In Table 1.3, where the mean is 138 and $s$ is 2.6, this means that 68% of the values fall within $138 \pm 2.6$, that is between 135.4 and 140.6, and 95% of the values fall within $138 \pm 5.2$, that is between 132.8 and 143.2.

**Significance**

5% significance means that if an investigation was carried out 100 times and each time there was a difference, then 95 of those differences are probably due to the factor being investigated and only 5 are probably due to chance.

## 1.2 The *t*-test

In order to decide whether the difference between two sets of data is important, or **significant**, we use the ***t*-test**. It compares the mean and standard deviation of the two sets of samples to see if they are the same or different.

A value for $t$ is calculated using a statistical formula. We then look up this value in a standard table of $t$-values, like the one in Table 1.4. Note that $t$, unlike standard deviation, does not have units. You do not need to

# 2 Cells

## Introduction

In the middle of the 17th century, one of the pioneers of microscopy, Robert Hooke (1635–1703), decided to examine a piece of cork tissue with his home-built microscope. He saw numerous box-shaped structures that he thought resembled 'monks' cells' or rooms, so he called them 'cells'. As microscopes became more sophisticated, other scientists observed cells and found that they occurred in every organism. No organism has yet been discovered that does not have at least one cell. Living things may vary in shape and size but scientists agree that they are all composed of cells. The study of cells has enabled us to learn more about how whole organisms function.

## 2.1 Cell theory

### Assessment statements

- Outline the cell theory.
- Discuss the evidence for the cell theory.
- State that unicellular organisms carry out all the functions of life.
- Compare the relative sizes of molecules, cell membrane thickness, viruses, bacteria, organelles and cells, using the appropriate SI unit.
- Calculate the linear magnification of drawings and the actual size of specimens in images of known magnification.
- Explain the importance of the surface area to volume ratio as a factor limiting cell size.
- State that multicellular organisms show emergent characteristics.
- Explain that cells in multicellular organisms differentiate to carry out specialised functions by expressing some of their genes but not others.
- State that stem cells retain the capacity to divide and have the ability to differentiate along different pathways.
- Outline one therapeutic use of stem cells.

### The cell theory

Today, scientists agree that the cell is the fundamental unit of all life forms. **Cell theory** proposes that all organisms are composed of one or more cells and, furthermore, that cells are the smallest units of life. That is, an individual cell can perform all the functions of life. Conversely, anything that is not made of cells, such as the viruses, cannot be considered living.

One of the functions carried out by all living organisms is reproduction. Therefore, the first principle of the cell theory is that cells can only come from pre-existing cells. They cannot be created from non-living material. Louis Pasteur (1822–1895) provided evidence for this. He showed that bacteria could not grow in a sealed, sterilised container of chicken soup. Only when living bacteria were introduced would more cells appear in the soup.

# 3 The chemistry of life

## Introduction

Living things are built up of many chemical elements, the majority of which are bonded together in organic, carbon-containing compounds. Most organic compounds in living things are carbohydrates, proteins, nucleic acids or lipids. Other inorganic, non-carbon-containing substances are also important but are present in much smaller quantities.

## 3.1 Chemical elements and water

### Assessment statements

- State that the most frequently occurring chemical elements in living things are carbon, hydrogen, oxygen and nitrogen.
- State that a variety of other elements are needed by living organisms, including sulfur, calcium, phosphorus, iron and sodium.
- State one role for each of the above elements.
- Draw and label a diagram showing the structure of water molecules to show their polarity and hydrogen bond formation.
- Outline the thermal, cohesive and solvent properties of water.
- Explain the relationship between the properties of water and its uses in living organisms as a coolant, medium for metabolic reactions and transport medium.

### Elements in living things

Carbon, hydrogen, oxygen and nitrogen are the four most common elements found in living organisms.

Carbon, hydrogen and oxygen are found in all the key **organic** molecules – proteins, carbohydrates, nucleic acids and lipids. Proteins and nucleic acids also contain nitrogen.

Any compound that does not contain carbon is said to be **inorganic**. A variety of inorganic substances are found in living things and are vital to both the structure and functioning of different organisms. Some important roles of inorganic elements are shown in Table **3.1**.

### Structure and properties of water

Water is the main component of living things. Most human cells are approximately 80% water. Water provides the environment in which the biochemical reactions of life can occur. It also takes part in and is produced by many reactions. Two of its most important properties (its solvent properties and its heat capacity) are due to its molecular structure, which consists of two hydrogen atoms each bonded to an oxygen atom by a covalent bond (Figure **3.1**).

The water molecule is unusual because it has a small positive charge on the two hydrogen atoms and a small negative charge on the oxygen atom.

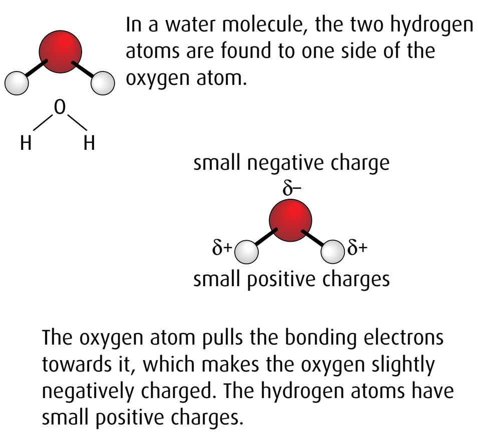



**Figure 3.1** The structure of a water molecule.

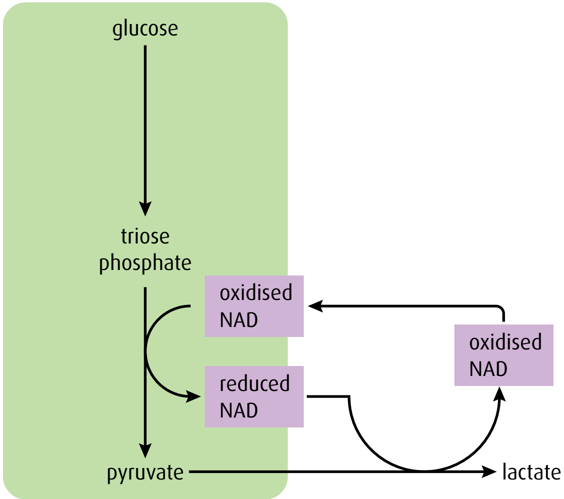



**Figure 3.20** Anaerobic respiration in animal cells.

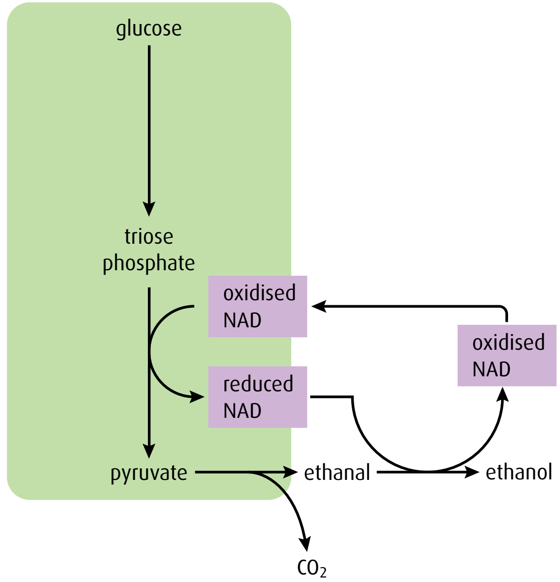



**Figure 3.21** Anaerobic respiration in yeast cells.

No further ATP is produced by the anaerobic respiration of pyruvate. Respiration is described in greater detail in Chapter **8**.

**12** What are the **two** products of anaerobic respiration in muscles?

**13** Where does aerobic respiration take place in a eukaryotic cell?

**14** Where in a cell does glycolysis occur?

## 3.8 Photosynthesis

### Assessment statements

- State that photosynthesis involves the conversion of light energy into chemical energy.
- State the light from the Sun is composed of a range of wavelengths (colours).
- State that chlorophyll is the main photosynthetic pigment.
- Outline the differences in absorption of red, blue and green light by chlorophyll.
- State that light energy is used to produce ATP, and to split water molecules (photolysis) to form oxygen and hydrogen.
- State that ATP and hydrogen (derived from the photolysis of water) are used to fix carbon dioxide to make organic molecules.
- Explain that the rate of photosynthesis can be measured directly by the production of oxygen or the uptake of carbon dioxide, or indirectly by an increase in biomass.
- Outline the effects of temperature, light intensity and carbon dioxide concentration on the rate of photosynthesis.

Photosynthesis means 'making things with light'. Glucose is the molecule most commonly made.

### Photosynthesis and light

The Sun is the source of energy for almost all life on Earth. Light energy from the Sun is captured by plants and other photosynthetic organisms, and converted into stored chemical energy. The energy is stored in

# 8 Cell respiration and photosynthesis

**Notation in biochemical pathways**

Photosynthesis and respiration involve large organic molecules. For simplicity, these are often identified by the number of carbon atoms they contain. Thus glucose, $C_6H_{12}O_6$, is simplified to $C_6$. A biochemical pathway may be written as $C_3 \rightarrow C_3$ so that nothing appears to have happened but the number of oxygen and/or hydrogen atoms in the second $C_3$ molecule will have changed. Both photosynthesis and respiration have linear and cyclic components in their pathways. Many of the compounds involved are abbreviated to their initials.

## Introduction

Respiration and photosynthesis are two key biochemical pathways in ecosystems. Light energy from the Sun is trapped as chemical energy in photosynthesis and then the energy is transferred through food chains and released back to the environment as heat energy from respiration. The two pathways can be simply written as:

$$6CO_2 + 6H_2O + \text{energy} \underset{\text{respiration}}{\overset{\text{photosynthesis}}{\rightleftharpoons}} C_6H_{12}O_6 + 6O_2$$

There are a number of similarities between the two pathways, which will be examined in this chapter. The basics of photosynthesis and respiration are covered in Chapter **3**, and it would be useful to review this chapter before proceeding.

## 8.1 Cell respiration

### Assessment statements

- State that oxidation involves the loss of electrons from an element, whereas reduction involves a gain of electrons; and that oxidation frequently involves gaining oxygen or losing hydrogen, whereas reduction frequently involves losing oxygen or gaining hydrogen.
- Outline the process of glycolysis, including phosphorylation, lysis, oxidation and ATP formation.
- Draw and label a diagram showing the structure of a mitochondrion as seen in electron micrographs.
- Explain aerobic respiration, including the link reaction, the Krebs cycle, the role of NADH + $H^+$, the electron transport chain and the role of oxygen.
- Explain oxidative phosphorylation in terms of chemiosmosis.
- Explain the relationship between the structure of the mitochondrion and its function.

### Oxidation and reduction

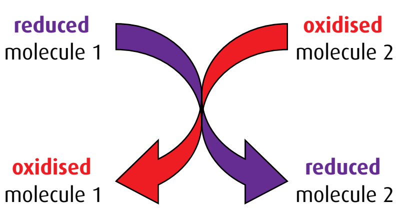



**Figure 8.1** Oxidation and reduction are linked processes – as one molecule is reduced another is oxidised in a redox reaction.

Cell respiration involves several **oxidation** and **reduction** reactions. Such reactions are common in biochemical pathways. When two molecules react, one of them starts in the oxidised state and becomes reduced, and the other starts in the reduced state and becomes oxidised, as shown in Figure **8.1**.

There are three different ways in which a molecule can be oxidised or reduced, as outlined in Table **8.1**. In biological oxidation reactions, addition of oxygen atoms is an alternative to removal of hydrogen atoms. Since a hydrogen atom consists of an electron and a proton, losing hydrogen atoms (oxidation) involves losing one or more electrons.

# Cells and energy  Option C

## Introduction

Photosynthesis and respiration are the key chemical reactions that enable ATP to be produced so that organisms can survive. All cells need energy to function and this comes from respiration. Respiration is a complex series of reactions, catalysed by enzymes, which release the chemical energy stored in sources such as glucose. Respiration provides cells with ATP that they can use to carry out their activities. Photosynthesis is the vital series of reactions by which plant cells build organic compounds, including glucose, from simple raw materials and light energy. An understanding of the biochemical events of respiration and photosynthesis, and how they are controlled, is crucial to understanding how cells use energy and stay alive.

## C1 Proteins

### Assessment statements

- Explain the four levels of protein structure, indicating the significance of each level.
- Outline the difference between fibrous and globular proteins with reference to two examples of each protein type.
- Explain the significance of polar and non-polar amino acids.
- State four functions of proteins, giving a named example of each.

### Protein structure

Proteins are large, complex molecules, usually made up of hundreds of amino acid subunits. The way these subunits fit together is highly specific to each type of protein, and is vital to its function. Figure **C.1** (overleaf) illustrates the structure of the protein hemoglobin.

The first stage of protein production is the assembly of a sequence of amino acid molecules that are linked by peptide bonds formed by condensation reactions. This sequence forms the **primary structure** of a protein. There are many different proteins and each one has different numbers and types of amino acids arranged in a different order, coded for by a cell's DNA.

The **secondary structure** of a protein is formed when the polypeptide chain takes up a permanent folded or twisted shape. Some polypeptides coil to produce an α helix, others fold to form β pleated sheets. The shapes are held in place by many weak hydrogen bonds. Depending on the sequence of amino acids, one section of a polypeptide may become an α helix while another takes up a β pleated form. This is the case in hemoglobin.

The **tertiary structure** of a protein forms as the molecule folds still further due to interactions between the R groups of the amino acids and within the polypeptide chain. The protein takes up a three-dimensional

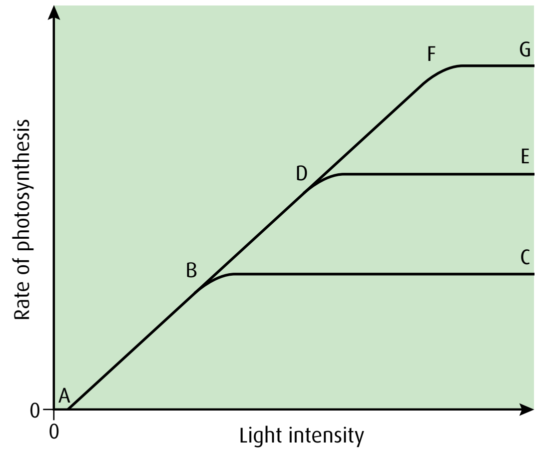



**Figure C.28** The effect of temperature and $CO_2$ concentration on the rate of photosynthesis.

- In Figure **C.28**, experiment 1 shows that as light intensity increases so does the rate of photosynthesis between the points A and B on the graph. At these light intensities, light is the limiting factor. Between points B and C another factor has become limiting.
- In experiment 2, where the temperature has been increased to 20 °C, the line has not changed, showing that it is the carbon dioxide concentration that is the limiting factor – a change in temperature has had no effect.
- In experiment 3, raising the carbon dioxide concentration causes the rate of photosynthesis to increase to point D, but then another limiting factor comes in to play from D to E.
- In experiment 4 we see that, at the higher carbon dioxide concentration, raising the temperature causes the line to rise to F but then again another limiting factor has become limiting from F to G.

The level of carbon dioxide in the atmosphere is relatively low and carbon dioxide is frequently a limiting factor for photosynthesis. Horticulturalists increase the yield of their crops by maximising the rate of photosynthesis in their glasshouses. They do this by keeping the air inside warm with heaters, increasing the intensity of light using lamps and, in some cases, increasing the concentration of carbon dioxide.

**28** If the graphs of the photosynthesis action spectrum and the chloroplast pigment absorption spectrum are superimposed what can be deduced?

**29** What is meant by the term 'limiting factor'?

**30** List **three** limiting factors for photosynthesis.

## End-of-chapter questions

1 Protein structure can be explained in terms of four levels – primary, secondary, tertiary and quaternary structure. Outline the quaternary structure of proteins. (3)

2 Explain the reasons for a large area of thylakoid membrane in the chloroplast. (2)

3 Explain why the chloroplast contains large amounts of RuBP carboxylase. (2)

4 Outline the difference between competitive and non-competitive enzyme inhibitors. (2)

5 Outline the differences between globular and fibrous proteins, giving a named example of each. (3)

6 Explain the significance of polar amino acids for membrane proteins. (2)

7 Explain how the proton gradient in the chloroplast is generated by chemiosmosis and what it is used for. (4)

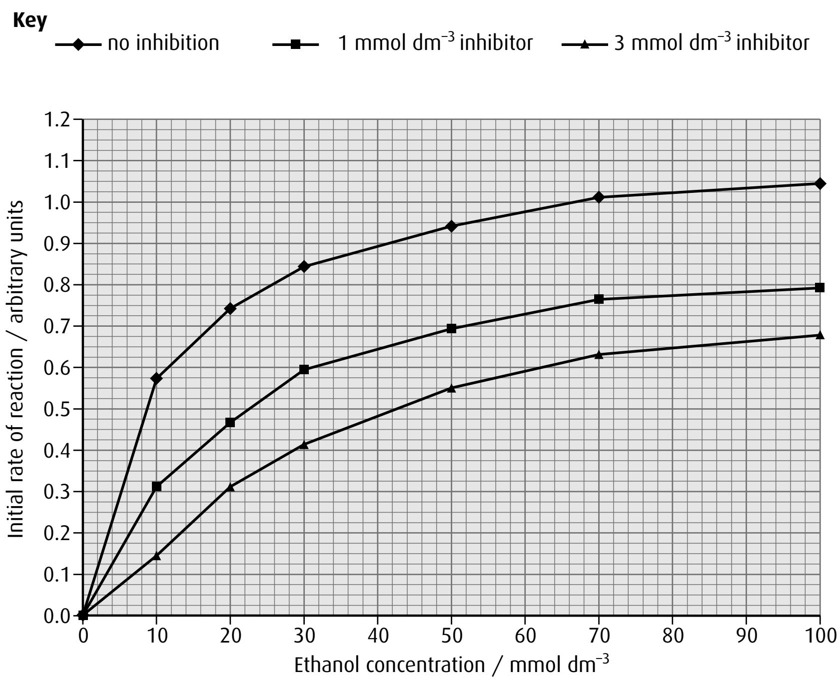
8 Explain why carbon dioxide concentration is a limiting factor of photosynthesis. (3)

9 Alcohol dehydrogenase is an enzyme that catalyses the reversible reaction of ethanol and ethanal according to the equation below.

$$\underset{}{NAD^+} + \underset{\text{ethanol}}{CH_3CH_2OH} \rightleftharpoons \underset{\text{ethanal}}{CH_3CHO} + NADH + H^+$$

The initial rate of reaction can be measured according to the time taken for NADH to be produced.

In an experiment, the initial rate at different concentrations of ethanol was recorded (no inhibition). The experiment was then repeated with the addition of 1 mmol $dm^{-3}$ 2,2,2-trifluoroethanol, a competitive inhibitor of the enzyme. A third experiment using a greater concentration of the same inhibitor (3 mmol $dm^{-3}$) was performed. The results for each experiment are shown in the graph below.




source: Taber, R (1998) *Biochemical Education*, **26**, pp 239–242

**a** Outline the effect of increasing the substrate concentration on the control reaction (no inhibition). (2)

**b** **i** State the initial rate of reaction at an ethanol concentration of 50 mmol $dm^{-3}$ in the presence of the inhibitor at the following concentrations:

1 mmol $dm^{-3}$

3 mmol $dm^{-3}$ (1)

**ii** State the effect of increasing the concentration of inhibitor on the initial rate of reaction. (1)

**c** Explain how a competitive inhibitor works. (3)

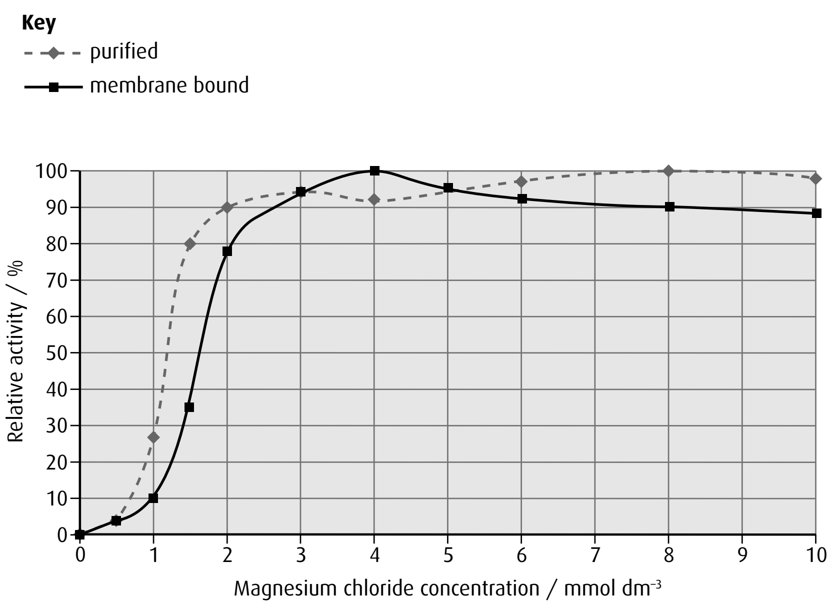
**(total 7 marks)**



**10** The hydrolysis of inorganic phosphate ($PP_i$) by phosphatase enzyme provides energy for a wide range of reactions. A phosphatase (PPase) occurs bound to thylakoid membranes. This enzyme was purified from the thylakoid membranes of spinach leaves using chromatography. The activity of the membrane-bound enzyme and the purified enzyme was measured.

The effect of the concentration of magnesium ions ($Mg^{2+}$) on the relative activity of these enzymes was determined using different concentrations of magnesium chloride. The concentration of inorganic phosphate used in both cases was of 1 mmol $dm^{-3}$.

| Activity of phosphatase/arbitrary units | |
|---|---|
| **Membrane bound** | **Purified** |
| 12 618 | 1215 |




reprinted from Po-Yin Cheung *et al.* (1998) 'Thiols Protect the Inhibition of Myocardial Aconitase by Peroxynitrite', *Archives of Biochemistry and Biophysics*, vol. 350, issue 1, pp. 104–108
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– they make it more likely that an organisms will survive and reproduce successfully in that environment. This is known as **convergent evolution**. An example is the wing of a bat and the wing of an insect (Figure **D.13**). The wings serve similar functions but they are derived from completely different structures. Bats and insects are not closely related.

Figure **D.14** summarises the differences between convergent and divergent evolution.


**Figure D.13** The bat's wing is structurally very different from the insect's wing, yet they perform a similar function. This is an example of convergent evolution.

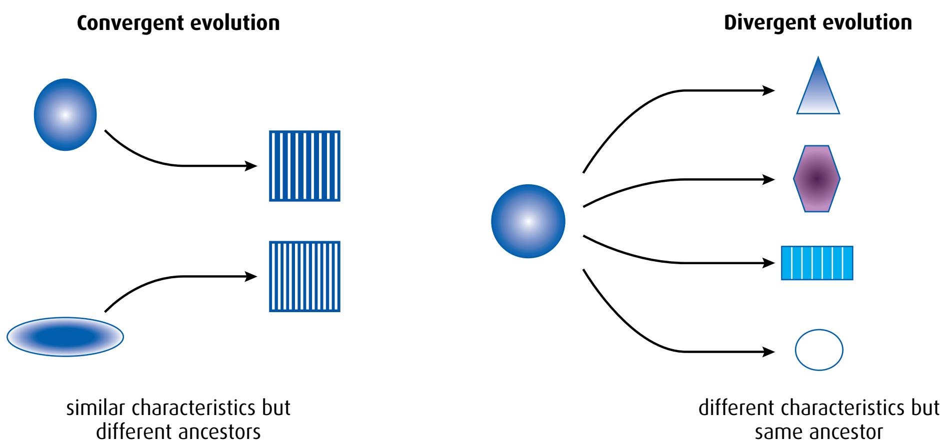


**Figure D.14** These diagrams show the difference between convergent and divergent evolution.

## The pace of evolution: gradualism and punctuated equilibrium

Darwin viewed evolution as a slow, steady process called **gradualism**, whereby changes slowly accumulated over many generations and led to speciation. For many species, this seems to be true. A good example of gradualism is the evolution of the horse's limbs, which fossils indicate took around 43 million years to change from the ancestral form to the modern one (Figure **D.15**, overleaf).

In some cases, the fossil record does not contain any intermediate stages between one species and another. A suggested explanation is that fossilisation is such a rare event that the intermediate fossils have simply not been discovered. In 1972, Stephen J Gould (1941–2002) and Niles Eldredge (b. 1943) suggested that the fossils had not been found because

| Hominid species | Lived | Features of head, skull and brain | Reconstructed image of skull | Other features |
|---|---|---|---|---|
| *Ardipithecus ramidus* | 4.4–4.3 mya (million years ago) in Ethiopia, East Africa | • foramen magnum is more ventral, indicating a trend towards bipedalism (Figure **D.19**)<br>• brain size: unknown | few fossils of this species have been found so the description is uncertain but it seems to have been quite similar to a chimpanzee with some important hominid features | • teeth seem to be intermediate between apes and Australopithecines – incisors are small and canines are smaller and more blunt than in apes, but molars are large and ape-like |
| *Australopithecus afarensis*<br>the most well-known specimen of this species is 'Lucy' | 3.9–2.9 mya in East Africa | • ape-like face with flat nose and protruding jaws<br>• large, tall lower jaw<br>• brain size: 375–550 $cm^3$ |  | • large molar teeth similar to *Ardipithecus ramidus* |
| *Australopithecus africanus* | 3.3–2.5 mya in southern Africa | • slightly flatter face but still with large, tall lower jaw<br>• brain size: 420–500 $cm^3$ |  | • smaller canines but molars still large |
| *Homo habilis*<br>first discovered in Olduvai Gorge in Tanzania; *H. habilis* used simple tools | 2.5–1.9 mya in eastern and southern Africa | • face more flattened than *Australopithecus*<br>• smaller lower jaw<br>• brain size: 500–800 $cm^3$ |  | • smaller teeth<br>• hips form distinct pelvic bowl |
| *Homo erectus*<br>specimens of *H. erectus* have been found in Europe and Asia as well as Africa so must have migrated from Africa; *H. erectus* were the first hominids to use fire | 1.8–0.3 mya in Africa, Indonesia, Asia, Europe | • face further flattened<br>• skull more rounded with large brow ridges<br>• smaller lower jaw<br>• brain size: 850–1100 $cm^3$ |  | |
| *Homo neanderthalensis*<br>neanderthal man | 150 000–30 000 years ago in Europe, western Asia | • face further flattened but still with large brow ridges<br>• rounded skull but lower forehead<br>• brain size: 1200–1625 $cm^3$ |  | • large teeth and jaw muscles<br>• limbs short relative to torso<br>• may have interbred with *H. sapiens* in Europe |
| *Homo sapiens*<br>modern man | 130 000 years ago to present | • flat face with no brow ridges<br>• reinforced lower jaw producing a chin<br>• rounded skull with high forehead<br>• brain size: 1200–1500 $cm^3$ |  | • smaller molars<br>• skeleton is less robust than other ancestors |

**Table D.2** Some examples of hominid species, from the fossil record.

# D5 Phylogeny and systematics

## Assessment statements

- Outline the value of classifying organisms.
- Explain the biochemical evidence provided by the universality of DNA and protein structures for the common ancestry of living organisms.
- Explain how variations in specific molecules can indicate phylogeny.
- Discuss how biochemical variations can be used as an evolutionary clock.
- Define 'clade' and 'cladistics'.
- Distinguish, with examples, between 'analogous' and 'homologous' characteristics.
- Outline the methods used to construct cladograms and the conclusions that can be drawn from them.
- Construct a simple cladogram.
- Analyse cladograms in terms of phylogenetic relationships.
- Discuss the relationship between cladograms and the classification of living organisms.

Aristotle (384–322 BC) grouped organisms into plants and animals. He then divided plants into three groups based on stem shape, and animals also into three groups based on where they lived – on land, in water or in the air. A further subdivision was based on whether the animal had red blood or not. Aristotle also devised a binomial system but it was not based on evolutionary relationships.

## Classifying organisms

With several million species of organism already known, and more being discovered every day, some way of grouping and organising them is essential. Classification enables scientists to identify newly found or unknown organisms, to see how organisms are related in evolutionary terms and to make predictions about the characteristics that organisms in a group are likely to have or share.

The Greek philosopher Aristotle attempted to devise a classification system 2000 years ago and many other biologists since then have added to and modified his system. Our modern biological classification began with the work of Carolus Linnaeus (1707–1778) who grouped organisms based on shared characteristics and devised the binomial system of naming organisms by their genus and species. He presented this work in his book *Systema Naturae* (1735). A natural classification such as the one devised by Linnaeus is based on identification of homologous structures that indicate a common evolutionary history. If these characteristics are shared between organisms then it is likely that they are related.

There are four main reasons why organisms need to be classified:

- to impose order and organisation on our knowledge
- to give species a clear and universal name, because common names vary from place to place
- to identify evolutionary relationships – if two organisms share particular characteristics then it is likely that they are related to each other, and the more characteristics they share then the closer the relationship
- to predict characteristics – if members of a particular group share characteristics then it is likely that other newly discovered members of that group will have at least some of those same characteristics.

10 **a** 335 (±5) W (*units required*) (1)
**b** as velocity increases power increases / positive correlation (1)
**c** both have same / similar maximum power;
both show increased power at greater velocities;
power tends to be higher at each velocity for $V_{70}$;
there is a greater range of powers at $V_{70}$ between 2.0 to 2.5 m $s^{-1}$ (2 max)
**d** risk of heart attack / heart failure from strenuous exercise (1)
(total 5 marks)


11 **a** directly proportional / positive correlation / more exercise leads to higher cardiac output (1)
**b** **i** 82.5 (±3)% (1)
**ii** 5.25 / 5.3 (*units not needed*) (1)
**c** cardiac output to muscles / overall cardiac output increases more in top athletes / with training;
training / top athlete decreases cardiac output to rest of body;
training increases oxygen consumption in muscle / overall oxygen consumption during heavy exercise;
training does not change oxygen consumption by rest of body during heavy exercise;
training causes changes by increasing stroke volume / lung capacity (3 max)
(total 6 marks)


## Option C

1 two or more polypeptides held together to form a single protein;
same bonding as for tertiary structure;
two types of bond named – disulfide bridges / ionic / hydrogen;
addition of a non-polypeptide structure / prosthetic group;
to form conjugated protein;
named example of quaternary structure e.g. hemoglobin (has four polypeptides) (3 max)

2 more chlorophyll / photosynthetic pigments / photosystems; (1)
more light absorbed (1)

3 RuBP carboxylase catalyses addition of carbon dioxide to RuBP;
carbon fixation;
start of / first step of the Calvin cycle;
Calvin cycle produces TP / glucose (2 max)

4 competitive inhibitors are similar to substrate molecules, and non-competitive inhibitors are not;
competitive inhibitor attaches to the active site, non-competitive inhibitor does not bind to the active site;
non-competitive inhibitor changes enzyme shape, competitive inhibitor does not;
effect of competitive inhibitor is reversible (2 max)

5 fibrous proteins have a long and narrow shape, globular proteins have rounded shape;
fibrous proteins are mostly insoluble in water, globular proteins are soluble in water;
fibrous is mainly secondary structure, globular at least tertiary structure;
*Examples include*:
collagen / actin / silk / keratin / other fibrous protein;
hemoglobin / myoglobin / any named enzyme / other globular protein (3 max)

6 polar amino acids are hydrophilic / 'water loving';
polar amino acids form hydrophilic channels in channel proteins;
allow hydrophilic / polar / charged particle substances to pass through the membrane;
control location of protein / hold the protein in the membrane;
polar amino acids on the surface of proteins make them water soluble (2 max)
*Accept any of the above points if clearly explained using a suitable diagram.*

7 electrons flow through proteins / electron transfer chain in thylakoid membrane;
electron flow causes protons to be pumped into the thylakoid space;
ATP synthase / synthetase located in the thylakoid membrane / next to electron transfer chain;
protons move from the thylakoid to the stroma through ATP synthase / synthetase;
down the concentration gradient;
ATP synthesised from ADP + $P_i$ (4 max)

8 rate of photosynthesis rises as $CO_2$ concentration rises;
up to a maximum when rate levels off / labelled graph showing outline of relationship;
when line levels off other factors become limiting;
$CO_2$ concentration in atmosphere is low;
$CO_2$ uptake reduced when stomata close to prevent water loss (3 max)

9 **a** directly proportional / greater concentration, greater rate of reaction;
at high concentrations the increase is smaller / plateaus / levels off (at approximately 70 mmol $dm^{-3}$) (2)
**b** **i** 1 mmol $dm^{-3}$: 0.70 (±0.02);
3 mmol $dm^{-3}$: 0.55 (±0.02); (1)
*Both needed for 1 mark. For 1 mmol* $dm^{-3}$ *accept 0.7.*
**ii** lower reaction rate at inhibitor concentration of 3 mmol $dm^{-3}$ / the greater the inhibitor concentration the slower the rate of reaction;
trend / overall shape are the same / increases but then levels off;
but lower at greater concentration of inhibitor (1 max)
**c** substrate and inhibitor (structurally) similar;
inhibitor binds to active site;
prevents substrate from binding;
activity of enzyme prevented;